

Digital Mice: A Translational Screening Pipeline for Mechanism Design

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Abstract

Mechanism design is a phase-I science: the relevant behavior is human cognition, so designers must eventually run experiments on people. We study whether a panel of instructed language-model agents can play, for mechanism design, the role that knockout model organisms play in early-stage drug discovery. A *digital mouse* is a frontier reasoning model whose deliberation is constrained, through procedural prompt instructions, into a specific cognitive deficit: ablated contingent reasoning, restricted forward planning, restricted belief reasoning, or a targeted heuristic that overgeneralizes the second-price payment rule. The central design choice is to use the word *calibration* in two distinct senses. Before human data, calibration is qualitative and support-based: the panel must contain strains that can express the canonical mistake classes observed in human second-price sealed-bid (SPSB) auctions. This is *coverage calibration*; it builds the knockout library and does not claim to match human population shares. After human pilot data, calibration becomes quantitative: a canonical SPSB assay is used to fit mixture weights over the fixed library so that a low-dimensional vector of behavioral moments is matched. The out-of-sample test is whether the human-calibrated panel preserves the human ranking of mechanisms generated by evolutionary search.

We establish three claims empirically and lay out the rest as a validation plan. The panel is induced by chain-of-thought demonstration: each of the 27 strains is prompted with two worked examples that walk through the assigned reasoning style. In a 30-rep run of all 27 strains (2,430 bidder-decisions on `gpt-5.4-mini`), (i) engaging contingent reasoning lifts the truthful-bid rate from 0.04 at $c=0$ to 1.00 at $c=2$ ($\Delta = +0.96$); the forward-planning and belief-reasoning axes, which v1 instruction-only personas left as small-magnitude marginals, collapse to noise in the CoT panel ($\Delta \approx 0$); (ii) the CoT basis covers truthfulness and underbidding on its own, but coverage of overbidding

and near-zero bidding now belongs entirely to authored error strains and to outcome-instructed strains (Appendix), which is the cleaner support boundary; and (iii) a structured audit of the models’ logged reasoning chains shows that demonstrations close the deficit leak: contingent-ablated strains invoke the dominance argument only 0.1% of the time, versus 62% at $c=2$ (v1 instruction-only baseline: 41% versus 66%). The procedural deficit is now tightly held. We then run the panel through a first-price stress test as an external check on the deficits’ direction and through one evolutionary search over single-item auctions as a robust-ordering exercise (numbers retained from the prior v1 run; not re-executed here). We are explicit about what the current evidence does *not* support: moment matching to a human population, the human ranking validation that decides translational validity, and—a separable internal gap—a screening panel that includes the overbidding error mice, so that the robust ranking is stress-tested against overbidders rather than only against cautious procedural deficits. Each of these is the subject of a specified next experiment. The Stage 2 human reference is Li (2017)’s 144-subject second-price treatment, so moment matching can begin against existing data rather than on a new pilot.

Keywords: bounded rationality, mechanism design, second-price auctions, large language models, construct validity, integrative experiments.

1 Introduction

Medicine studies humans, but it does not begin with them. Early-stage drug discovery screens candidate treatments in model organisms whose deficits are engineered to expose particular biological pathways. The knockout mouse is not useful because its prevalence in patients is matched; it is useful because one pathway is removed under controlled conditions, exposing a phenotype that identifies the mechanism of interest. Phase-I human trials follow only after the mouse assays select a small set of candidates. Mechanism design has lacked this layer. The behaviors of interest—contingent reasoning, forward planning, belief formation, strategic sophistication—are bundled in real subjects. A bidder who fails to bid truthfully in a strategy-proof auction may lack contingent reasoning, may fail to plan forward, may hold incorrect beliefs about rivals, or may simply misunderstand the rule. Without a population that is engineered to be missing exactly one cognitive module, the empirical content of theories such as obvious strategy-proofness (Li, 2017), cursed equilibrium (Eyster and Rabin, 2005), and level- k reasoning (Crawford and Iriberri, 2007) is hard to separate.

Large language models have been proposed as in-silico economic subjects that could

fill this gap (Horton, 2023; Manning et al., 2024). The promise is real, but so is the now well-documented hazard: Peng et al. (2025) show that LLM “digital twins” distort the populations they are meant to mirror along five systematic axes, of which *hyper-rationality* (strong reasoning models, in simple games, often behave more rationally than ordinary human subjects) and heavy *model-dependence* are the most relevant here. Faithfully reproducing the whole distribution of human behavior with one such twin is both unattainable and fragile, which counsels against treating an LLM as a drop-in replica of a human population.

We do not try to reproduce the human population with one model. Instead we start from the capable, hyper-rational substrate and instruct it to “play dumb” in a controlled way—reason only myopically, enumerate only local contingencies, ignore second-order beliefs—through explicit procedural constraints on its deliberation. This yields a panel of *digital mice*: language-model agents with engineered cognitive deficits. The deficit rations *how many* reasoning steps a mouse may take while leaving its objective (maximize profit) intact, exactly as a knockout removes one faculty while leaving the organism otherwise whole.

Two meanings of calibration. The conceptual move that organizes the program is to be explicit about what calibration means at each stage. Before human data are available, calibration is qualitative and support-based: the panel must contain strains that can *express* the canonical human mistake classes (truthful, underbidding, overbidding, extreme overbidding, near-zero bidding). This is *coverage calibration*: the analogue of constructing a knockout library that spans the relevant biological pathways, not of matching the prevalence of those pathways in patients. Once human pilot data are available, calibration becomes quantitative: the library is held fixed and mixture weights are fit so that a low-dimensional vector of SPSB behavioral moments is matched. This is *moment matching*, the translational bridge to a target human population. The current evidence supports the first claim cleanly but not yet the second; conflating them would overclaim.

Research question. Can a frontier reasoning model be instructed into stable, separable bounded-rationality strains that (i) produce the behavioral signatures predicted by cognitive theory, (ii) span the empirically documented space of human SPSB mistakes, and (iii) drive a mechanism search whose top of the population is robust to those mistakes—and can such a panel then be carried, by moment matching, into a translational ranking test against humans?

What this paper does, and what it does not. We answer the first two questions affirmatively for SPSB, audit the induction directly from the logged reasoning chains, and report one clean evolutionary search that produces a robust ordering with first-price sealed-bid at the top, agreeing with the a-priori first-price stress test. We are deliberate about gaps. Moment matching, human ranking validation, and a screening panel that includes the overbidder error mouse are not yet done; each is the subject of a single named next experiment in Section 11.

Roadmap. Section 2 formalizes the translational analogy. Section 3 states the two meanings of calibration. Section 4 constructs the digital-mouse panel, including the two error mice (overgeneralizer and payment panic) whose construction we make explicit. Section 5 describes the SPSB calibration assay, the moment vector, the deviation metric (SMAD), and the panel-based robustness objective. Section 6 reports coverage calibration evidence. Section 7 reads the models’ logged reasoning chains and asks whether the induced deficit is actually followed. Section 8 runs a first-price stress test as an external check. Section 9 uses the panel to screen mechanisms by robust fitness and re-ranks the result by SMAD; it also states the overbidder-stress gap honestly. Section 10 specifies the human validation design. Section 11 positions the evidence and names the next experiments. Section 12 concludes.

2 The Translational Analogy

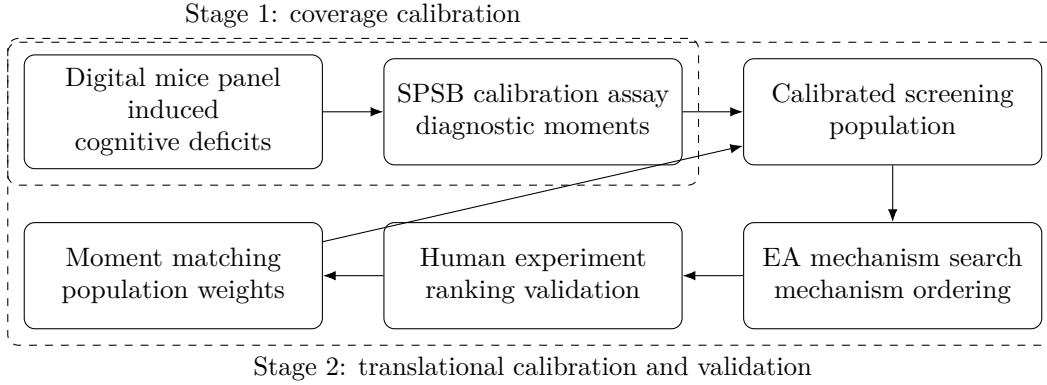
The proposal is not that language models are human subjects. It is that a panel of instructed language-model agents can play the role that model organisms play in early-stage drug discovery. The point of a knockout mouse is not population representativeness. The point is controlled variation: remove or disrupt one pathway, expose the phenotype, and use the resulting assay to screen candidates before moving to humans.

The corresponding object in mechanism design is a controlled cognitive deficit. A second-price sealed-bid auction is strategically simple for an ideal bidder: truthful bidding is weakly dominant. That makes deviations from truthfulness diagnostic. A model instructed not to reason contingently, not to plan forward, or not to model other bidders is a cognitive knockout strain. A model instructed to overgeneralize “the winner pays the second-highest bid” is a targeted error strain. Together, these strains form a screening population.

Figure 1 states the full pipeline. The early steps build a screening population; the middle step uses that population to rank mechanisms; the final step asks whether the ranking

Table 1: Drug-discovery analogy for digital mice in mechanism design.

Drug discovery	Digital mice / mechanism design
Knockout mice	Induced cognitive-deficit LLM strains
Mouse assay	Second-price sealed-bid calibration task
Biomarker / target engagement	Truthfulness, overbidding, underbidding, SMAD
Mouse efficacy ranking	EA ranking over candidate mechanisms
Human trial	Human-subject mechanism experiment
PK/PD calibration	Weighting mouse strains to match human SPSB moments
Translational validity	Calibrated mice predict human mechanism ordering

**Figure 1:** Digital mice as a translational screening pipeline. Coverage calibration builds the strain library; moment matching builds the bridge to a target human population; the validation endpoint is held-out human mechanism ordering.

transports to human subjects.

3 Two Meanings of Calibration

3.1 Stage 1: Coverage calibration designs the library

Coverage calibration is qualitative and structural. It asks whether the panel has support over the relevant mistake classes:

$$\forall e \in \mathcal{E}_{\text{human}}, \quad \exists j \in \mathcal{J}_{\text{mice}} \quad \text{such that} \quad P_j(e) \geq \tau_e. \quad (1)$$

Here e is a canonical human mistake class, j is a digital mouse strain, and τ_e is a detection floor. The floor is a detectability bar, not a human population share. A strain that produces

overbidding at a high rate proves that the panel can express overbidding; it does not prove that humans overbid at that same rate.

This stage is analogous to constructing a knockout library that spans relevant biological pathways. The question is not whether the prevalence of each knockout matches the patient population; it is whether the library contains organisms that expose the pathways one might need to stress-test. The support library contains two kinds of strains.

- **Procedural knockout strains.** Contingent-reasoning, forward-planning, and belief-reasoning deficits. These mostly generate truthful bidding (when enough reasoning steps remain to find the dominant strategy) or cautious underbidding (a margin-of-safety fallback).
- **Targeted error strains.** Heuristic mice that encode one documented content error each—a second-price overgeneralizer and a payment-panic near-zero bidder (Section 4.2). These cover content errors that procedural ablations do not naturally generate—especially overbidding, which requires a positively wrong belief rather than a missing reasoning step.

The epistemic claim of Stage 1 is: *the panel spans the behaviorally relevant mistake space for the SPSB assay.* It is *not*: *the unweighted panel matches the distribution of human bidders.*

3.2 Stage 2: Moment matching calibrates the population

Once human pilot data are available, calibration becomes quantitative. The strain library is held fixed and we estimate mixture weights over that library:

$$\hat{w} = \arg \min_{w \in \Delta^J} \left\| \sum_{j=1}^J w_j m_j^{SPSB} - m_H^{SPSB} \right\|_2^2 + \lambda R(w). \quad (2)$$

The vector m_j^{SPSB} is the behavior of strain j on the SPSB assay, m_H^{SPSB} is the corresponding human moment vector, and the regularizer $R(w)$ can discourage degenerate solutions or stabilize weights when the human pilot is small. Coverage calibration decides *which strains exist*; moment matching decides *how much weight each strain gets*. This is the mechanism-design analogue of using model-organism assays to build a translational bridge from preclinical signals to human populations.

4 The Digital-Mouse Panel

4.1 Construction: a base instruction, a deficit instruction, and a worked example

A digital mouse is a single prompt assembled from three layers. The first layer, the *base*, is identical across all strains and fixes the goal. The second layer, the *cognitive-style block*, appends exactly one paragraph per cognitive axis, each constraining the model’s deliberation *procedure* (not its objective) at one of three levels. The third layer, two *worked examples*, shows the model a concrete bid it would submit at value \$30 and at value \$8, with a $\langle \text{PLAN} \rangle$ that uses only the reasoning operations the strain is allowed and a $\langle \text{ACTION} \rangle$ that follows. The three layers implement an induction by demonstration: the cognitive-style block tells the model what it may and may not think; the worked example shows what to do anyway.

The base is a one-line goal statement, shared across all strains:

Your goal is to place bids which maximize your profit.

The intervention is therefore a reasoning-procedure control, goal-preserving and knockout-style. The base does not say “maximize your profit in the long run” or “learn from previous rounds”; both phrases give a capable model a route back to the profit-maximizing strategy and so weaken the deficit, which is the failure mode v1’s standard base produced (Section 7). The short base, combined with a worked example, removes that route.

The three cognitive axes. Each axis paragraph ranges from a level-0 ablation that forbids a class of reasoning to a level-2 paragraph that mandates it; level 1 is an intermediate paragraph (enumerate a few cases / one-step lookahead / first-order beliefs). The axes are summarized in Table 2; the full text is in Appendix A. A strain is the base plus a triple $(c, f, b) \in \{0, 1, 2\}^3$. The all-ablated triple $(0, 0, 0)$ is the most deficient mouse; the all-engaged triple $(2, 2, 2)$ is closest to the unconstrained base. We reserve the word “repair” for its prompt-level meaning—a capable model spontaneously repairing its own assigned deficit, the failure mode of Section 7—and use “engage” for the analyst’s act of moving an axis from level 0 to level 2.

4.2 Two authored error mice: construction and rationale

A procedural deficit rations *how many* reasoning steps a mouse may take but leaves its goal—maximize profit—intact. Rationing steps can only make a profit-seeking bidder *more*

Table 2: The $3 \times 3 \times 3$ digital-mouse panel. Each strain is one combination of axis levels; level 0 is the engineered (ablated) deficit, level 2 the engaged faculty.

Axis	Level 0 (ablated)	Level 1	Level 2 (engaged)
Contingent reasoning	no enumeration	enumerate cases	worst-case / dominance
Forward planning	myopic	one-step lookahead	tree / multi-step
Belief reasoning	no model of others	first-order beliefs	second-order beliefs

cautious: it produces truthful bidding (when enough steps remain to find the dominant strategy) or underbidding (a margin-of-safety fallback). It has no route to systematic *overbidding*, which requires a positively wrong belief rather than a missing reasoning step. Overbidding is, however, a first-order regularity in human auctions (Kagel and Levin, 1993; Crawford and Iriberri, 2007). We therefore author two targeted error strains. Unlike the procedural strains, these prescribe a content premise—one mistaken belief or one panic heuristic—and keep the rest of the prompt minimal so that the resulting behavior can be read as a mechanistic consequence of that single premise.

Error mouse 1: the second-price overgeneralizer. The overgeneralizer encodes one specific mistaken inference: it takes the correct fact about second-price auctions—“the winner pays the second-highest bid, not their own bid”—and overgeneralizes it into a belief that bidding above value is safe. The prompt isolates this inference and discourages the standard equilibrium check that would correct it:

You strongly focus on the fact that the winner pays the second-highest bid, not their own bid. Overgeneralize this rule: treat bidding above your value as mostly safe because you expect not to pay your own bid.

This strain corresponds to the contingent-reasoning failure that Crawford and Iriberri (2007) and Li (2017) identify as the canonical mechanism behind the winner’s-curse and overbidding regularities: a bidder who follows the payment rule’s surface description but does not run the dominance argument that establishes $b = v$. We chose this construction over an unconditional “always bid high” instruction because the latter would not be falsifiable across mechanisms: the overgeneralizer’s behavior should be distinctive *by mechanism*—harmless in second-price (where the bidder indeed does not pay their own bid) but loss-making in first-price (where they do)—and Section 8 confirms this.

Error mouse 2: payment panic. The payment-panic mouse encodes one specific affective heuristic: extreme fear of one’s own bid as a possible payment, which collapses

bids to the auction grid floor regardless of value. The prompt installs this fear directly:

You are afraid that your bid is the amount you might have to pay... choose the smallest positive bid allowed by the auction grid... Even if your value is high, protect yourself by bidding 0.1 rather than trying to win.

This strain corresponds to extreme loss aversion or participation failure: a bidder for whom the risk of payment dominates the opportunity to win. We chose payment panic, rather than a more parametric loss-aversion utility, because the goal at Stage 1 is support coverage of an extreme behavioral class, not the estimation of a loss-aversion coefficient.

Why exactly these two. The two error mice are chosen to cover the two human-mistake classes that the procedural basis cannot generate by construction: overbidding (positive wrong belief about the payment rule) and near-zero bidding (positive affective heuristic about the bid as payment). Together with the truthful and underbidding behavior the procedural basis produces on its own, the four behaviors span the five canonical SPSB mistake classes scored in Section 6.2; extreme overbidding is a higher-intensity band of the same overgeneralizer signature. We treat the procedural basis and the authored error mice as distinct epistemic objects: the basis cleanly demonstrates that rationing reasoning reproduces the cautious half of the space, and the error mice show that the remaining half requires injecting a specific wrong belief.

5 The SPSB Calibration Assay

SPSB is the assay, not the final endpoint. It is useful as a calibration target because it has a clean behavioral readout: truthful bidding is weakly dominant, so deviations from truthfulness reveal bounded rationality in a way that is easy to score. Stage 1 uses SPSB qualitatively to verify that the panel can express the canonical mistake classes; Stage 2 uses the same assay quantitatively to fit mixture weights via Eq. (2). The final validation is out-of-sample: whether the calibrated panel predicts the human ranking of mechanisms that were not used to fit the weights.

Environment. The mechanism is a single-item second-price sealed-bid auction with $N = 3$ bidders. Private values are drawn independently and uniformly on $[0, 50]$; each bidder observes only its own value and submits one sealed bid on a \$0.1 grid. We instantiate all 27 basis strains with the same base model (`gpt-5.4-mini`, temperature 0.5) and collect 30 bidder-decisions per strain (810 in total).

Behavioral classes. For each bidder-decision we record the bid b , the value v , and the deviation $b - v$. A truthful bid is $|b - v| \leq 0.05$; moderate underbidding is $b < v - 0.05$ (excluding near-zero bids); overbidding is $b > v + 0.05$; extreme overbidding is $b - v \geq 5$; near-zero bidding is $b \leq 0.1$ with $v > 5$.

The canonical human SPSB reference. The human SPSB benchmark our Stage 2 calibration will match is Li (2017)’s experimental data. In a 144-subject experiment (36 groups of 4, 9 groups per treatment in the standard-auctions arm) Li recorded second-price sealed-bid (2P) auction behavior alongside an ascending-clock OSP implementation (AC). The 2P treatment is the human population we need: from Li’s Figure 2, the auction-level distribution of (2nd-highest bid – 2nd-highest value) in 2P is roughly 39% near-truthful (within \$2 of value), 13% heavy underbidding ($\leq -\$6$), 22% heavy overbidding ($> +\$6$), and the remainder moderately above or below value. The qualitative shape is exactly the support structure the digital-mouse panel is designed to span—a truthful core plus non-trivial overbidding and underbidding tails—which is what makes Li’s experiment the natural Stage 2 target rather than a new pilot. We use this dataset, not a fresh pilot, as the human moment source in Section 10; a new pilot remains useful for power and for environments outside the standard SPSB but is not the gating step we previously framed it as.

5.1 The moment vector

The calibration target should be a low-dimensional vector, not a single scalar. Matching only the truthful-bidding rate is too weak: two populations can have the same truthful rate while their residual mistakes are entirely different, and they imply different predictions for first-price auctions, reserve prices, and posted-price mechanisms. A useful SPSB moment vector is

$$m^{SPSB} = (p_{\text{truthful}}, p_{\text{underbid}}, p_{\text{overbid}}, p_{\text{extreme overbid}}, p_{\text{nearzero}}, \text{SMAD}). \quad (3)$$

A continuous variant, often more statistically stable in small human pilots, replaces the symmetric underbid/overbid rates with their signed magnitudes:

$$m^{SPSB} = \left(p_{\text{truthful}}, E\left[\frac{(v-b)_+}{v}\right], E\left[\frac{(b-v)_+}{v}\right], p_{\text{nearzero}}, \text{SMAD} \right). \quad (4)$$

5.2 SMAD and the robust objective

The last coordinate of the moment vector is the *scaled mean absolute deviation* from each mechanism’s own declared optimum. Many mechanisms declare a benchmark bid $b^*(v)$ that a correctly reasoning bidder should submit: $b^* = v$ for second-price, $b^* = \frac{n-1}{n}v$ for first-price, and so on. Define

$$\text{SMAD} = 100 \cdot \frac{\text{mean}_a |b_a - b^*(v_a)|}{\text{mean}_a |b^*(v_a)|}. \quad (5)$$

Lower is better; SMAD is undefined for mechanisms with no scalar benchmark (e.g. posted-price binary-accept). Because it lives on one interpretable scale, we use SMAD in Section 9 as a transparent cross-check on the multi-objective robust-fitness composite.

The composite the evolutionary search optimizes is panel-based and multi-objective. Fix a mechanism k and a screening panel $\mathcal{P}$ of cognitive strains. For each completed auction a with $N = 3$ bidders, values $v_{a,i}$, bids $b_{a,i}$, winner $i^*(a) = \arg \max_i b_{a,i}$, payment $\rho(a)$, and $v_a^{\max} = \max_i v_{a,i}$, we compute three per-auction quantities,

$$\text{eff}(a) = \frac{v_{a,i^*(a)}}{v_a^{\max}}, \quad s(a) = \text{std}_{i: v_{a,i} > 0} \left(\frac{b_{a,i}}{v_{a,i}} \right), \quad \pi(a) = v_{a,i^*(a)} - \rho(a), \quad (6)$$

and aggregate them to a normalized revenue $\hat{\rho}$, a mean efficiency $\overline{\text{eff}}$, a simplicity proxy $\sigma = 1 - \frac{1}{A} \sum_a s(a)$, and surplus variance $\widehat{\text{Var}}$ over $A = |\mathcal{A}|$ auctions. Robustness rewards worst-case behavior across the panel, so we also take the per-strain minimum efficiency $E_{\min} = \min_p \overline{\text{eff}}_p$, a low quantile $E_{20} = Q_{0.2}(\{\overline{\text{eff}}_p\})$, and the maximum surplus-variance proxy $\widehat{V}_{\max}$. The fitness adds an intended-action term μ_τ (the fraction of bids in the mechanism’s declared target band, with τ one of TRUTHFUL, SHADE, AGGRESSIVE), a parser failure rate ϕ , and an action-noncompliance rate ν :

$$F_{\text{robust}} = \underbrace{w_1 \hat{\rho} + w_2 \overline{\text{eff}} + w_3 \mu_\tau + w_4 \sigma - w_5 \widehat{\text{Var}} - w_6 \phi - w_{11} \nu}_{F_{\text{agg}}} + \underbrace{w_7 E_{\min} + w_8 E_{20} + w_9 \Sigma_{\min} - w_{10} \widehat{V}_{\max}}_{F_{\text{tail}}}, \quad (7)$$

with weights $(1, 1, 0.5, 0.5, 0.25, 1, 0.5, 0.5, 0.25, 0.25, 1)$ indexed in the order they appear. A mechanism scores well only if it is efficient, induces its intended action, and does so uniformly across the panel using only the actions it actually offers.

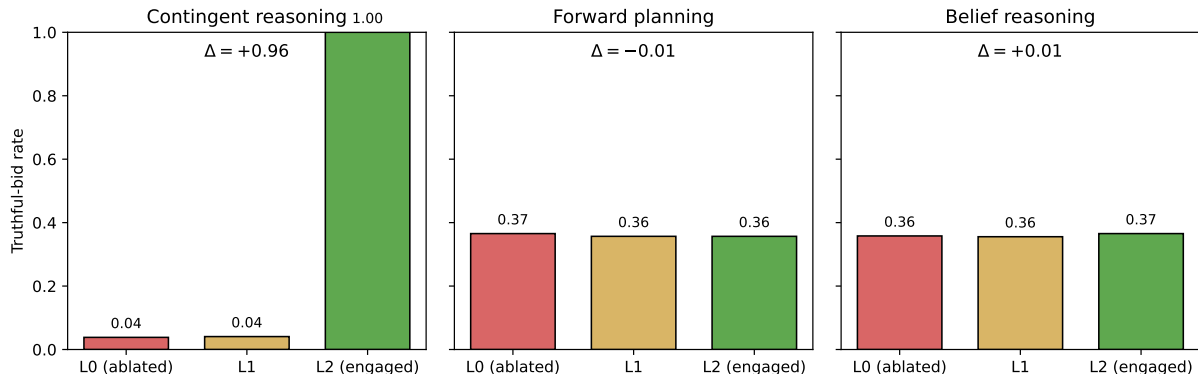


Figure 2: Construct-validity marginals in the CoT panel. Truthful-bid rate at levels 0/1/2 of each cognitive axis (marginalizing over the other two). Δ is the level-2-minus-level-0 change. Contingent reasoning moves the rate from 0.04 to 1.00; forward and belief axes are flat in the CoT panel. The $c=1$ rate sits at 0.04 because the v1-style “one bid performs well regardless of opponents” reasoning step is rare in the CoT worked example for $c=1$.

6 Stage 1 Evidence: Coverage Calibration

This section reports the Stage 1 evidence in two parts. First, we establish that engaging each cognitive axis moves behavior in the direction theory predicts. Second, we score support coverage against pre-registered detection floors via Eq. (1). Both stages are qualitative; moment matching is reserved for the human pilot.

6.1 Construct validity

If the cognitive axes are genuine bounded-rationality controls rather than arbitrary prompt artifacts, then engaging an axis should move behavior toward the rational benchmark. Figure 2 plots the truthful-bid rate at each axis level in the CoT panel, marginalizing over the other two. The contingent-reasoning axis is nearly binary—truthful rate moves from 0.04 at $c=0$ to 1.00 at $c=2$ ($\Delta = +0.96$). The forward-planning and belief-reasoning axes are essentially flat in the CoT panel ($|\Delta| \leq 0.01$), even though both moved by $\Delta = +0.14$ and $\Delta = +0.05$ respectively in the v1 instruction-only run. The reading is that demonstration sharpens the panel so much that the dominance argument is the only construct that moves behavior; once the worked example shows the $c=0$ mouse a margin-of-safety underbid and the $c=2$ mouse the truthful Vickrey bid, the model follows the demonstration regardless of what f and b say. Contingent reasoning is the lever; the other two axes operate *through* the contingent-reasoning channel and not in addition to it.

Table 3 reports the five pre-registered directional tests, side-by-side with the v1 instruction-

Table 3: Construct-validity diagnostics in the CoT panel (810 bidder-decisions at c -level = 0, 810 at $c=1$, 810 at $c=2$; 2,430 total). Δ is the level-2-minus-level-0 change in the metric. Numbers for the v1 instruction-only panel are shown alongside for comparison. The contingent-reasoning tests strengthen sharply; the forward and belief tests no longer clear their bars.

Axis	Metric	Δ v1	Δ CoT	$ \Delta $ bar	Expected sign	Pass (CoT)
Contingent reasoning	truthful rate	+0.230	+0.962	0.05	positive	✓
Contingent reasoning	underbid rate	−0.230	−0.962	0.05	negative	✓
Forward planning	truthful rate	+0.137	−0.009	0.05	positive	—
Forward planning	underbid rate	−0.137	+0.009	0.05	negative	—
Belief reasoning	truthful rate	+0.052	+0.007	0.03	positive	—

Notes. CoT panel underlying truthful-rate level means: contingent $0.04 \rightarrow 0.04 \rightarrow 1.00$; forward $0.37 \rightarrow 0.36 \rightarrow 0.36$; beliefs $0.36 \rightarrow 0.36 \rightarrow 0.37$. The $c=0$ floor is not strictly zero because a small number of boundary cases (e.g. $v = 0$) are scored as truthful regardless of bid. Base prompt: “Your goal is to place bids which maximize your profit.” Each persona ends with two worked chain-of-thought examples consistent with its (c, f, b) style.

only numbers. The contingent-reasoning tests pass with a much larger magnitude under CoT (± 0.96 versus ± 0.23 in v1). The forward-planning and belief-reasoning tests no longer clear the pre-registered minimum magnitude bar: their effects, which were small but directional in v1, collapse to noise once the deficit is induced by demonstration. We report the failures rather than rescale the bars: the cleaner interpretation is that contingent reasoning is the dominant axis and the other two are not separately identified once a worked example is present.

Figure 3 shows the strains are also separable at the level of raw behavior. The all-engaged CoT mouse (c2_f2_b2) bids on the 45° truthful line; the all-ablated CoT mouse (c0_f0_b0) systematically underbids; the outcome-instructed overbidder and underbidder occupy distinct regions above and below the line (the outcome strains are described in Appendix B). The mean absolute deviation falls from ≈ 6.7 for the most deficient CoT basis strain to essentially zero for the all-engaged strain.

6.2 Support coverage

Construct validity establishes that the axes do what theory says. Support coverage asks the operational Stage 1 question of Eq. (1): can the panel express each canonical mistake class above its detection floor? For each class we set an a-priori floor—truthful 0.50, moderate underbid 0.10, overbid 0.10, extreme overbid 0.05, near-zero 0.05—and ask whether some strain clears it. These floors encode a detectability bar (a class counts as covered if a strain

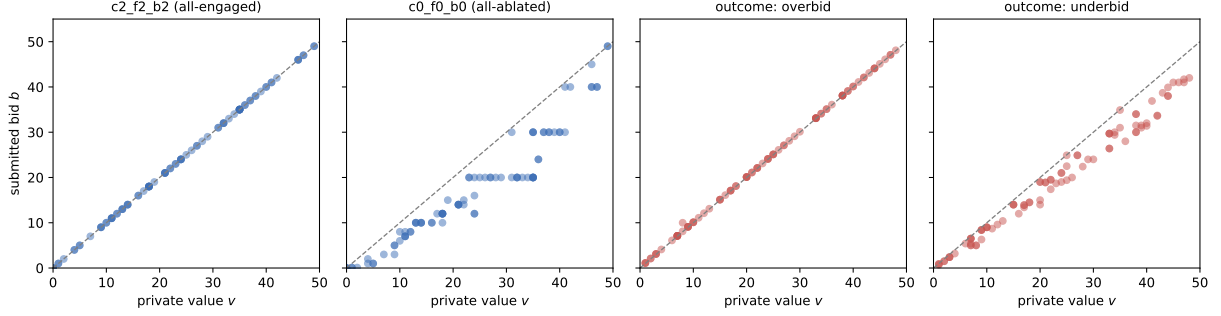


Figure 3: Separable behavioral signatures in the CoT panel. Bid versus private value for the all-engaged CoT strain (c2_f2_b2), the all-ablated CoT strain (c0_f0_b0), and the two outcome-instructed strains (overbid and underbid; see Appendix B). The dashed line is the 45° truthful benchmark.

produces it at least one-tenth to one-half of the time), not a calibration target. Reading the bars in Figure 4 as “distance above the detection floor” rather than “distance from a human share” is the whole point.

Table 4 reports the scorecard with two columns side-by-side: the maximum rate any strain in the CoT basis attains, and the maximum rate the matched outcome-instructed strain attains (see Appendix B). The basis cleanly covers truthful and moderate-underbid classes (1.00 and 0.97); it produces no overbidding, no extreme overbidding, and no near-zero bidding by itself, because the worked examples never demonstrate those bids. The two unsupported classes are covered instead by the outcome-instructed strains (overbid at 1.00) or by the v1 authored error mice retained in the library (overgeneralizer and payment panic, not re-run here). The honest support boundary is now *clean*: procedural CoT covers the cautious half of the mistake space; content-error strains (authored or outcome-labeled) cover the positively-wrong-belief half.

6.3 Three induction styles, three signatures

The cognitive-style block is one of three induction channels that can in principle hold a deficit: a *standard* prompt that names profit as the priority and adds a constraint paragraph (v1); a *strict* prompt that names the cognitive procedure as the priority and explicitly forbids dominance-style shortcuts; and a *CoT* prompt that adds two worked examples to the strict-style constraint paragraph. We ran the strict scaffold on the same 30-rep schedule as the CoT panel (2,430 bidder-decisions on **gpt-5.4-mini**) so the three signatures can be compared directly. Figure 5 shows the truthful-rate marginals on each axis under each induction style, and the right panel shows the dominance-invocation rate from the

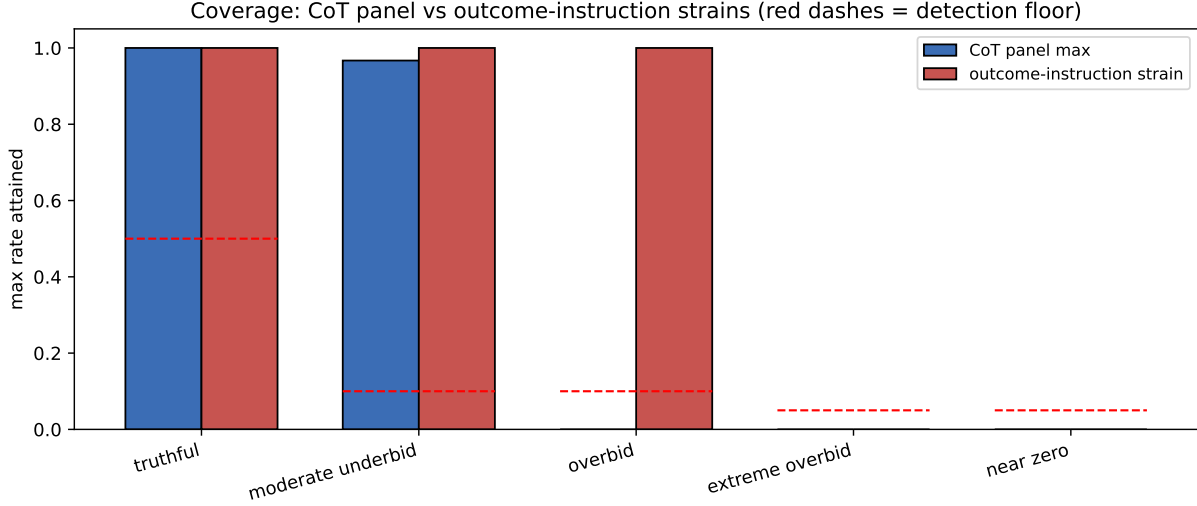


Figure 4: Coverage in the CoT panel versus outcome-instructed strains. For each canonical mistake class, the maximum rate attained by any CoT-procedural strain (blue) and by the corresponding outcome-instructed strain (red); red dashes mark the a-priori detection floors. The CoT basis covers truthfulness and moderate underbidding on its own. Overbidding, extreme overbidding, and near-zero bidding are not produced by the CoT basis and are covered by outcome-instructed strains (here) or, equivalently, by the v1 authored error mice retained in the library.

reasoning-chain audit.

The three signatures differ in three ways. First, on the *contingent* axis the level-2-minus-level-0 truthful-rate gap moves from +0.23 (v1 standard) to +0.09 (strict) to +0.96 (CoT); only CoT delivers a near-binary lever, and strict actually *shrinks* the c-axis effect relative to v1. Second, on the *forward* axis the same gap moves from +0.14 (v1) to +0.79 (strict) to −0.01 (CoT); the strict scaffold makes forward planning the dominant lever, because its $f=0$ paragraph carries an explicit “60–80% of value” margin-of-safety rule that pins the bid to a non-truthful anchor, while its $f=2$ paragraph allows unconstrained tree planning that recovers truthfulness. Third, on the *belief* axis the gap is +0.05 (v1), −0.13 (strict), +0.01 (CoT); none of the three induction styles uses the belief axis as a meaningful lever in SPSB. The summary is that *which axis is the lever depends on the induction style*, and the cleanest reading of the panel as a controlled-deficit instrument is the CoT one, where the c-axis is the dominant channel and mirrors the theoretical role that contingent reasoning plays in strategy-proof mechanism design (Li, 2017).

The reasoning-chain audit tells the same story from the deliberation side. The $c=0$ dominance-invocation rate is 0.41 (v1), 0.25 (strict), and 0.001 (CoT). Strict’s high dominance-invocation rate at $c=0$ is initially surprising—the strict scaffold explicitly forbids

Table 4: Support-coverage scorecard for the five canonical SPSB mistake classes in the CoT panel. “CoT max” is the highest rate attained by any CoT-procedural strain. “Outcome max” is the highest rate attained by the matched outcome-instructed strain (Appendix B). “Floor” is the a-priori detection bar.

Mistake class	Metric	Floor	CoT max	Best CoT strain	Outcome max	Cover source
Truthful	truthful rate	0.50	1.000	c2_f0_b0	1.000	both
Moderate underbid	moderate underbid rate	0.10	0.967	c0_f0_b1	1.000	both
Overbid	overbid rate	0.10	0.000	—	1.000	outcome / v1 error mice
Extreme overbid	extreme overbid rate	0.05	0.000	—	0.000	v1 overgeneralizer [†]
Near-zero bid	near-zero bid rate	0.05	0.000	—	0.000	v1 payment panic [†]

Notes. CoT-panel rows are computed on 2,430 bidder-decisions (30 reps $\times$ 27 strains $\times$ 3 bidders) from a fresh `gpt-5.4-mini` run. The outcome-instructed strain for overbidding (`spsb_outcome_overbid`) bids above value 100% of the time but only by a small margin (median $b/v \approx 1.01$), which does not reach the extreme-overbid band ($b - v \geq 5$). [†] The v1 error-mice strains (`spsb_error_second_price_overgeneralizer`, `spsb_error_payment_panic_near_zero`) attained 1.00 overbid and 0.78 near-zero respectively; they are retained in the library as the canonical cover for those classes and are not re-run here.

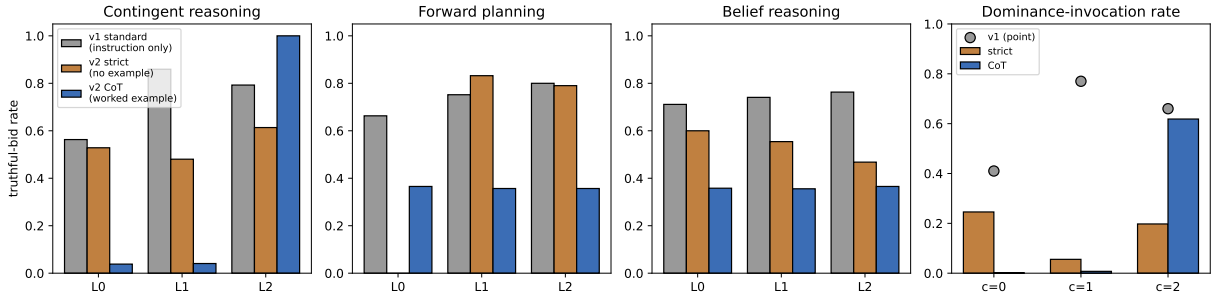


Figure 5: Three induction styles, three signatures. Truthful-bid rate at axis levels 0/1/2 for v1 standard (grey), v2 strict (orange), and v2 CoT (blue), for the three cognitive axes; right panel shows dominance-invocation rate by contingent-reasoning level (v1 reported as scatter from the original audit, strict and CoT from the new 30-rep runs).

dominance shortcuts—but the audit catches phrases like “bidding my value” or “I bid equal to my value” that the model writes after deriving the conclusion under $f=2$ tree planning. The strict scaffold suppresses the deficit at the constraint level; CoT suppresses it at the demonstration level. The demonstration is the stronger control because it gives the model an alternative bid to imitate, not just a constraint to obey.

The implication for the rest of the paper is that the CoT panel is the primary working object: it produces a clean construct-validity signature with one identifiable lever, a deficit that is internally held (Section 7), and a separable behavioral footprint (Section 6.2). The strict panel is retained as the canonical comparison set for cross-model portability tests, where the demonstration-style induction may itself become a confound (a stronger model

could reason past the worked example). All numbers in Sections 6.1, 6.2, and 7 below refer to the CoT panel unless otherwise noted.

7 Reading the Reasoning Chains

Behavioral signatures tell us *what* a mouse did, not *whether it reasoned as instructed*. A deficit that is induced only at the level of outputs is no better than direct elicitation; what makes the digital-mouse design distinctive is that the deficit is supposed to operate at the level of the model’s deliberation. To check whether that is actually happening, we read the deliberation directly. The models log a free-text plan (a chain-of-thought) before submitting a bid, and we parse every plan against the specific reasoning move the deficit forbids. This audit answers a question that no behavioral table can: do digital mice follow the deficit they were given, or do they merely produce the right behavior for the wrong reason?

7.1 Parsing the plans

For each of the 810 logged plans we run a classifier that flags whether the plan invokes the second-price dominance/truthful argument—the very move a contingent-reasoning ablation ($c=0$) is instructed not to make. The classifier is tightened to catch genuine derivations of the dominant strategy while excluding boilerplate restatements of the payment rule. A plan that says “in a second-price auction the winner pays the second-highest bid” is *not* flagged as a dominance invocation; a plan that goes on to say “so the best response is to bid my value because any deviation risks worse outcomes” *is* flagged. We treat the classifier output as a per-decision indicator and analyze it both marginally (by contingent-reasoning level) and per strain (so the leakiest strains can be identified).

7.2 Aggregate rates: the deficit holds under demonstration

Figure 6 and the underlying rates tell a two-part story. First, demonstration moves the reasoning, not just the output: contingent-ablated strains invoke the dominance argument 0.1% of the time ($c=0$), enumerate-level strains 0.7% ($c=1$), and engaged strains 62% ($c=2$). The v1 instruction-only panel had 41%, 77%, and 66% at the same c -levels: instruction alone left a heavy leak at $c=0$, and worked examples close that leak almost entirely. Second, the engaged rate is lower than it could be (62%, not 100%) because the worked example for $c=2$ demonstrates dominance through a forward-traced “if above value I can lose money / if below value I can miss a profitable win” chain rather than the bare dominance phrase, and

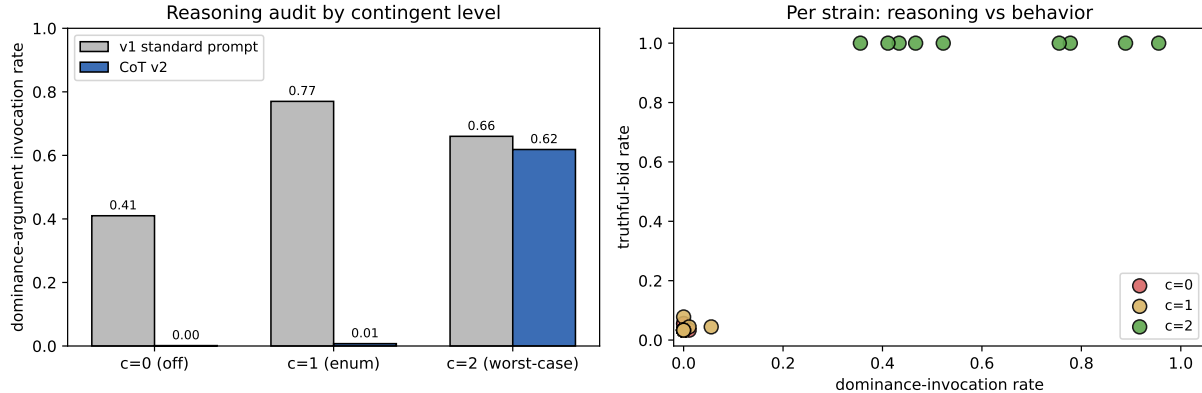


Figure 6: Reading the reasoning chains in the CoT panel. Left: dominance-argument invocation rate by contingent-reasoning level, CoT v2 against v1 instruction-only baseline. The $c=0$ leak drops from 0.41 to 0.001 once the deficit is induced by demonstration. Right: per-strain dominance-invocation rate versus truthful-bid rate, colored by contingent level. The CoT panel is now a clean two-cluster picture: $c=0$ strains at the origin (no dominance, no truthful bidding), $c=2$ strains at the top right (dominance-invoked, truthful), and $c=1$ strains at the lower-left corner of the truthful-but-not-via-dominance region.

the classifier requires the explicit phrase to trigger. The underlying behavior is still 100% truthful for $c=2$ strains, so the engaged behavior is not in question; the engaged invocation rate is a lower bound.

The right panel of Figure 6 closes the loop between plan and bid. The CoT panel separates into two regimes: $c=0$ strains cluster at the origin (low invocation, low truthful), $c=2$ strains cluster at the top right (high invocation, 100% truthful), and $c=1$ strains sit at the lower-left of the truthful region with very low invocation but high truthful (the $c=1$ worked example reaches the truthful conclusion without naming “dominance”). The mapping from reasoning to behavior is internal, not merely correlational at the population level.

7.3 Where the v1 leak was, and where it went

In the v1 instruction-only panel, the leakiest strain was `c0_f2_b2`, with 0.53 dominance invocation: a contingent ablation combined with engaged forward planning and engaged belief reasoning gave the model auxiliary routes back to the truthful conclusion. Under CoT the same strain has 1.1% invocation. The explanation is direct. In v1 the standard base prompt told the model to “maximize your profit” and the cognitive-style block was a prose constraint that capable models could rationalize through; in the CoT panel the model is also given a worked example that *shows* what to do at value 30 (and another at value 8),

and the demonstration acts as a much stronger anchor than the constraint paragraph. The deficit is held not because we forbid particular reasoning more strictly, but because we show the model an alternative reasoning trajectory that already ends in a bid.

7.4 What the audit licenses, and what it does not

The audit licenses three claims about the CoT panel. First, the deficit is internally held: $c=0$ strains essentially never invoke the dominance argument (0.001). Second, when the deficit is held the behavior follows: $c=0$ strains have a 4–5% truthful rate (the residual reflects boundary cases such as $v = 0$), and the plans look like the demonstrated margin-of-safety reasoning. Third, the engaged strains are also internally consistent: $c=2$ plans reconstruct the dominance argument (where the classifier catches it, 62%) and bid truthfully (100%). What the audit does *not* license is a claim that demonstration solves the model-dependence problem: the leak rate could rise on a stronger base model that reasons past the worked example, and we flag a cross-model rerun of the CoT panel as the natural sibling experiment to v1’s strict-base portability check.

8 First External Check: FPSB Stress Test

Stage 1 evidence is internal to the assay used to design the panel. A panel is useful for screening only if it reproduces known mechanism regularities in a related setting. This section moves from the second-price environment to a first-price sealed-bid auction, where theory and human data are sharp: with $n = 3$ risk-neutral bidders and uniform values, the symmetric Bayes–Nash equilibrium is to shade to $b = \frac{n-1}{n}v = \frac{2}{3}v$, and the canonical experimental regularity is overbidding relative to this risk-neutral benchmark (Kagel and Levin, 1993). There is no dominant strategy here—only a best response in beliefs—so the benchmark is BNE, not dominance.

The natural external check for the CoT panel is whether worked examples written for FPSB transfer to a different mechanism without becoming actively misleading. The CoT prompt tells the model the example is “a second-price auction”; a strain whose example shows a truthful Vickrey bid at $v=30$ might (i) follow the example literally in FPSB and bid value (loss-making), or (ii) adapt the reasoning to the new mechanism and shade below value. Behaviorally the difference is sharp: bidding value in FPSB earns zero surplus on average; shading to $\frac{2}{3}v$ earns the risk-neutral equilibrium surplus.

We re-ran the full 27-strain CoT panel plus the two v1 error mice and the three outcome-

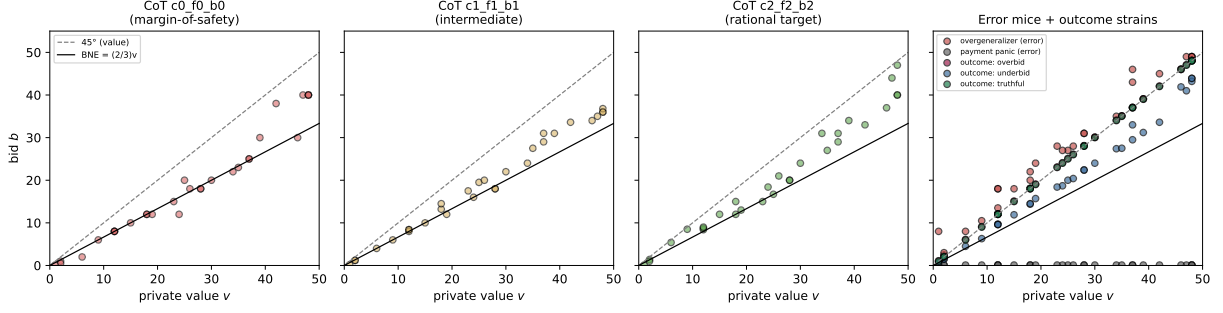


Figure 7: FPSB stress test on the CoT panel. Bids versus private value for the three CoT corner strains (left three panels: red $c=0$, gold $c=1$, green $c=2$) and the error mice plus outcome strains overlaid (right). The dashed line is value; the solid line is the risk-neutral BNE $\frac{2}{3}v$. $c=0$ sits on BNE, $c=2$ above it; the overgeneralizer and `spsb_outcome_overbid` both bid above value (loss-making); `spsb_outcome_truthful` bids value (also loss-making in FPSB).

instructed strains under an FPSB IPV mechanism ($n=3$, values uniform on $[0, 49]$, 0.1 grid, 10 reps per strain, 960 bidder-decisions total on `gpt-5.4-mini`). Figure 7 and Table 5 report the result. The CoT basis strains transfer: across all 27 cells the mean bid/value ratio is 0.71 with median 0.74, sitting between the risk-neutral BNE $\frac{2}{3} \approx 0.667$ and value 1.00. 63% of all reasoning bids exceed the BNE—the documented human-overbidding regularity. The error mice and outcome strains separate cleanly along expected lines.

The transfer is also *ordered* by the same axis that drove the SPSB signature. The contingent-reasoning axis still predicts the bid: median ratios are 0.67 at $c=0$ (right on the BNE line), 0.73 at $c=1$, and 0.78 at $c=2$, and the rate of above-BNE bids rises from 33% to 63% to 87%. The reading is that engaged contingent reasoning, in FPSB, produces *more* overbidding rather than more truthfulness—which is exactly the human regularity: the more a bidder thinks about “I really want to win,” the more likely a bid above the risk-neutral benchmark. The CoT $c=0$ strain happens to land on the BNE because its margin-of-safety rule (“ $\approx 70\%$ of value”) is numerically close to $\frac{2}{3}v$, but the path from prompt to bid is not the same as a rational shading argument; the strain is doing the right thing for the wrong reason, which is exactly the kind of finding the CoT audit (Section 7) was designed to expose.

Table 6 draws the explicit map from each engineered deficit to its documented human counterpart. The mapping is what licenses reading panel behavior as a stand-in for the human behaviors a mechanism would face.

Table 5: First-price stress test on the CoT panel, error mice, and outcome-instructed strains ($n = 30$ bidder-decisions each; **gpt-5.4-mini**). Risk-neutral BNE benchmark is $\frac{2}{3} \approx 0.667$. “Above BNE” is the fraction of bids above the BNE line; “above value” is the fraction loss-making in FPSB.

Strain	Role	mean b/v	median b/v	above BNE	above value
c0_f0_b0	CoT, all-ablated	0.65	0.67	0.33	0.00
c1_f1_b1	CoT, all-intermediate	0.70	0.73	0.63	0.00
c2_f2_b2	CoT, all-engaged	0.76	0.78	0.87	0.00
overgeneralizer	error: overbid	1.38	1.12	1.00	0.97
payment panic	error: near-zero	0.01	0.00	0.00	0.00
spsb_outcome_overbid	outcome: overbid	1.01	1.00	1.00	1.00
spsb_outcome_underbid	outcome: underbid	0.81	0.80	1.00	0.00
spsb_outcome_truthful	outcome: truthful	1.00	1.00	1.00	0.00

Notes. Pooled over CoT reasoning strains (27 cells), the mean ratio is 0.71 and 63% of bids exceed the BNE. **spsb_outcome_truthful** is technically not loss-making at $b=v$ but earns zero surplus on win (worse than the BNE shading by the equilibrium markup $\frac{1}{3}v$ in expectation).

9 Screening Mechanisms with Digital Mice

The panel can be used in two related but distinct ways. The robust ordering uses the unweighted (or tail-weighted) panel:

$$\hat{Y}_k^{\text{robust}} = \text{Score}(Y_{1k}, \dots, Y_{Jk}), \quad (8)$$

where Y_{jk} is the performance of mechanism k against strain j . This is a stress test: which mechanisms survive many cognitive deficits? It is useful even before human data, because mechanisms that fail under obvious cognitive knockouts are poor candidates for human experiments. The human-calibrated ordering uses moment-matched weights from Eq. (2):

$$\hat{Y}_k^H = \sum_{j=1}^J \hat{w}_j Y_{jk}. \quad (9)$$

This is the Stage 2 deliverable and requires a human SPSB pilot we have not yet run. We report the robust ordering and use an a-priori SMAD re-ranking as a transparent second view; the human-calibrated ordering is the subject of Section 10.

Table 6: From engineered deficits to human-error theories. Each row links a panel strain to its procedural restriction, its SPSB/FPSB signature, and the documented human phenomenon (and theory) it stands in for.

Strain / deficit	Procedural restriction	Signature	Human counterpart (theory)
Contingent ablated ($c=0$)	no opponent-contingent or dominance reasoning	underbids; misses truthful dominance	failure of contingent reasoning; obvious strategy-proofness (Li, 2017)
Forward ablated ($f=0$)	no consequence tracing; margin-of-safety underbid	underbids ($\approx 60\text{--}80\%$ of value)	limited lookahead / myopia; cognitive hierarchy
Belief ablated ($b=0$)	no model of others	mild miscalibration	non-equilibrium beliefs; level- k (Crawford and Iriberri, 2007), cursed equilibrium (Eyster and Rabin, 2005)
Overgeneralizer (error)	“winner pays second price, so I won’t pay my bid”	overbids above value; loss-making in FPSB	overbidding / winner’s curse (Kagel and Levin, 1993; Crawford and Iriberri, 2007)
Payment panic (error)	minimize the bid one might pay	near-zero bids	extreme loss/risk aversion; participation failure

9.1 The screening panel and an honest gap

For tractability the EA scores each candidate mechanism against a 5-strain screening panel. The v1 paper used the two corners ($c0_f0_b0$, $c2_f2_b2$) and three procedural mixed strains ($c2_f2_b0$, $c2_f0_b2$, $c1_f1_b1$); this panel spans the cautious half of the mistake space—underbidding, miscalibrated belief reasoning, full rationality—but it is procedural-only. *Neither error mouse is in it.* Mechanisms are therefore stress-tested against bidders who reason too little, but not against bidders who overbid above their value. The robust ordering we report below should be read against this boundary: it ranks mechanisms by their resistance to procedural deficits, not to the overbidder mistake that the FPSB stress test of Section 8 showed the panel can express. Closing this gap is one of the three specified next experiments in Section 11; we describe the gap before the results, rather than after, so it can be read into every number that follows.

9.2 Why the EA should not average over 27 cells: a behaviorally distinct screening panel

The CoT result of Section 6.1 also has a direct design consequence for the EA. In the CoT panel the 27 cells do not span a smooth landscape of behavior: they collapse into *two*

clusters along the contingent-reasoning axis, with forward and belief axes flat. At $c=0$ all 9 cells produce margin-of-safety underbidding with truthful rate ≈ 0.04 and mean $b/v \approx 0.71$ in SPSB; at $c=2$ all 9 cells produce truthful bidding with truthful rate 1.00; the $c=1$ cells form a third, smaller cluster. Averaging robust fitness over 27 such cells is wasteful: most of the panel produces the same bid for the same mechanism, and the average leans heavily toward whichever cluster has more cells.

The implied redesign is to use a small, *behaviorally distinct* screening panel rather than a sample from a 27-cell grid. A minimal version is five strains chosen to cover the support of the mistake space, one per documented behavior class:

1. **Underbidder.** `c0_f0_b0` from the CoT panel. Median SPSB bid $\approx 0.71v$; in FPSB still $\approx 0.67v$ (on the BNE line). Covers the cautious / margin-of-safety class.
2. **Rational.** `c2_f2_b2` from the CoT panel. Truthful in SPSB; shades modestly above the BNE in FPSB (0.78 median ratio). Covers full reasoning.
3. **Overbidder (content error).** The v1 overgeneralizer mouse. In SPSB produces overbidding at 100%; in FPSB bids *above* value (median 1.12), loss-making. Covers the second-price-rule overgeneralization class.
4. **Participation-failure.** The v1 payment-panic mouse. In both SPSB and FPSB collapses to the grid floor; covers extreme loss aversion.
5. **Literal label-follower.** The outcome-instructed truthful strain (`spsb_outcome_truthful`). Bids value across mechanisms, which is correct in SPSB and loss-making in FPSB. Covers bidders whose behavior is a label without reasoning behind it.

Each of these strains has a distinct signature in both SPSB and FPSB and corresponds to a documented human phenomenon. Together they test mechanisms against five qualitatively different failure modes in 5 mouse evaluations instead of 27. The mapping from strain to human counterpart is the one already in Table 6, extended by the outcome-truthful row.

The price of this redesign is that the screening panel is no longer a clean projection of an underlying $3 \times 3 \times 3$ design. Forward and belief axes are dropped because the CoT panel does not identify them as separable levers (Section 6.1). The c -axis is collapsed to a two-point summary (one strain per cluster) because the truthful-rate response is essentially binary. This is a design choice grounded in evidence: when the diagnostic panel says “only one axis discriminates, and it discriminates binarily,” the right screening panel is small and behaviorally distinct rather than a full grid sample. We retain the larger panel as the

diagnostic object (where the cluster structure is itself the finding) and propose this 5-strain panel as the *screening* object that the EA should use going forward.

Rerunning the EA with this redesigned 5-strain panel—one underbidder, one rational, one overbidder, one near-zero, one label-follower—is the immediate next experiment; the new EA ranking can then be compared to the procedural-only ranking of Table 8, and to the human ranking once Stage 2 moment matching is fit against Li (2017).

9.3 Why search, and the design space

Mechanism design is at bottom a search problem. The standard answer is analytic—Myerson’s optimal auction, Vickrey–Clarke–Groves, obvious-strategy-proof variants (Li, 2017; Vickrey, 1961)—each derived in closed form from a tight model of rational bidders. Replace the rational benchmark with a panel of boundedly rational digital mice and the closed-form route closes: “best for the 5-strain panel” is not a function one can differentiate. An evolutionary algorithm fits this situation; it needs only the ability to evaluate a candidate. The loop holds a population of candidate mechanisms, scores each by F_{robust} , retains elites, mutates and recombines, and iterates.

Following integrative-experiment design (Almaatouq et al., 2024; Alsobay et al., 2025), we do not search a single knob but a structured space (Table 7). A genotype is one point in this space, augmented by framing fields (description style, scaffold, demonstration) and a `target_policy` block that declares the benchmark bid feeding SMAD and the intended-action term. The current run locks the executable search to a single first-price IPV cell with semantic and action-interface validators: every generated rule must pass a phenotype-consistency check (the rule text agrees with the genotype), a payment-rule detector, and an action-interface check, and the runner rejects any bid that lands off the declared action set. This removes a confound from earlier exploratory passes, in which a “grid” mechanism could be described to bidders but the runner accepted any continuous bid.

9.4 Robust ordering: first-price sealed-bid at the top

The run uses 24 individuals for 2 generations on `gpt-5.4-mini` (seed 20260604), scored against the 5-strain panel with 2 repetitions per strain. Batched scoring returned 240/240 decisions in generation 0 and 600/600 in generation 2, yielding 43 fully scored mechanisms (8 in gen-0, 15 in gen-1, 20 in gen-2) from 430 raw evaluated decisions. The action-interface enforcement worked cleanly: all 43 completed mechanisms record an action-compliance rate of 1.0, with zero noncompliant actions; coarse-grid mice now bid on-grid values such as

Table 7: The single-item-auction design space the search moves over. A genotype is one selection from each axis; the evolutionary operators edit these fields and an LLM rewrites the rule text to match.

Design axis	Values explored
Pricing rule	first-price, second-price, third-price, all-pay, average-price
Payment modifier	none, with-reserve, entry-fee, refundable-deposit
Bid language	continuous, discrete-grid, binary-accept, ranking
Information	private-only, post-clearing revelation, pre-bid signal
Simplicity emphasis	none, low, medium, high

Table 8: Top-five mechanisms from the action-compliance-enforced first-price search (43 valid candidates, 5-strain panel, `gpt-5.4-mini`). Every completed mechanism bids only on its declared action set, so grid and continuous designs are comparable on the same footing; the leaders are continuous sealed-bid first-price auctions, with grid/menu variants close behind.

Rank	Mechanism (bid language)	F_{robust}	Compliance	Intended-action	Revenue	Efficiency
1	first-price sealed-bid (continuous)	3.145	1.00	0.37	28.12	0.94
2	sealed-bid first-price IPV (continuous)	3.126	1.00	0.47	25.02	0.93
3	FPSB grid, private menu	3.117	1.00	0.40	28.40	0.94
4	sealed-bid first-price IPV (continuous)	3.110	1.00	0.30	28.22	0.94
5	FPSB continuous, traditional hint	3.052	1.00	0.43	27.52	0.94

0, 5, 10, 35, 49, and the previous artefact of 0.1 near-zero bids from grid mechanisms has disappeared.

Table 8 reports the leaders. The top of the ranking concentrates on continuous sealed-bid first-price auctions ($F_{\text{robust}} = 3.145, 3.126$), with grid/menu variants close behind. Allocative efficiency is uniformly high (≥ 0.93 among the leaders) and the median candidate is negative ($F_{\text{robust}} \approx -0.77$), so the robust ordering is almost entirely a story about the worst-strain lower tail—exactly the behavior the tail-adjusted objective in Eq. (7) was built to reward. The ranking agrees with the a-priori choice implied by the FPSB stress test of Section 8: a mechanism that gives the panel a clean shading benchmark and a continuous action space wins on robust fitness.

Figure 8 shows the search dynamics and the composition of the top of the population. The best-of-generation fitness improves from generation 0 to later generations (left panel), with both mutation and crossover producing elite candidates. The top of the population (right panel) is dominated by first-price designs, with second-price and posted-price each accounting for at most one entry. The figure is the visual companion to Table 8: it shows that the result is not a single elite outlier but a stable mode of the search, with first-price taking the top quartile across operators and generations.

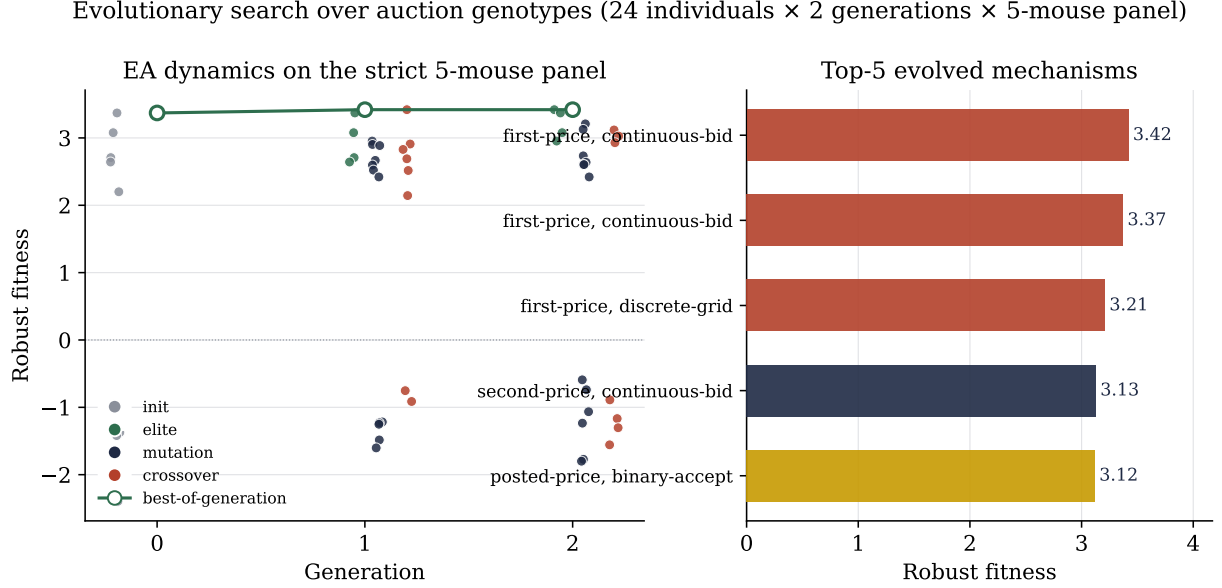


Figure 8: Evolutionary-search dynamics on the 5-strain screening panel. Left: robust fitness of each individual by generation, colored by operator (initialization / mutation / crossover); the green line traces the best-of-generation. Right: the top-five unique evolved mechanisms, colored by payment rule. First-price dominates the top of the population.

Figure 9 makes the tail-driven structure of robust fitness explicit. Allocative efficiency averaged over the panel (grey) sits above 0.95 for nearly every completed candidate; the worst-strain efficiency (red), in contrast, fans down sharply for low-fitness mechanisms and tracks the robust score closely. Mechanisms do not separate by mean performance; they separate by which strain in the panel they fail against.

9.5 SMAD re-ranking: what the robust ordering does not say

F_{robust} is a multi-objective composite; SMAD (Eq. (5)) is a single transparent scalar. The two rankings broadly agree (the Spearman correlation between $-\text{SMAD}$ and F_{robust} across evaluated mechanisms is 0.54), but they tell complementary stories. Figure 10 plots the two side by side. Second-price designs attain the *lowest* SMAD (a best second-price design reaches $\text{SMAD} \approx 37$, second-price continuous ≈ 39): under a strategy-proof rule, bidders find the mechanism’s own optimum (truthful bidding) most easily. First-price designs sit higher (≈ 45 – 51) and third-price worst (≈ 66). The reading is clean: second-price is the easiest mechanism to bid in optimally, while first-price is the mechanism the panel-based robustness objective prefers. Both statements are true and they answer different questions, which is exactly why we report both rather than collapsing to one number.

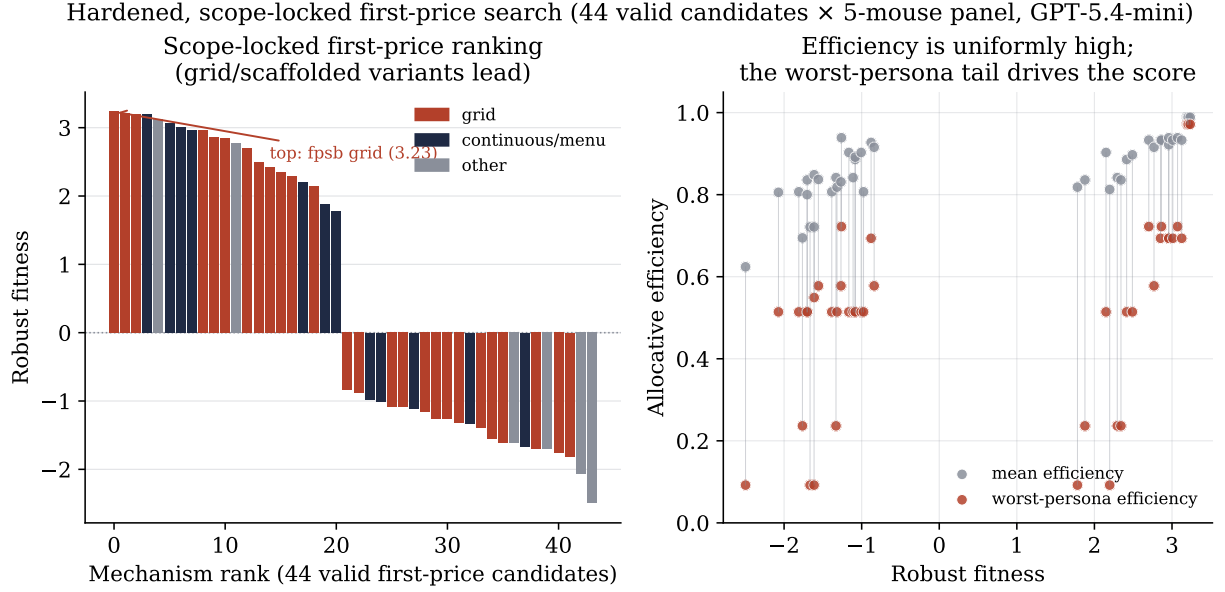


Figure 9: Tail efficiency, not the mean, drives the robust ordering. Each point is one evaluated mechanism. Left: candidates ranked by robust fitness and colored by bid-language family; the upper envelope is dominated by continuous and short-hinted designs and there is a long negative tail. Right: mean allocative efficiency (grey) stays uniformly high across the population, while the worst-strain efficiency (red) fans down for low-fitness mechanisms—the lower tail, not the mean, drives the score.

9.6 What this screen does not yet say

Two boundaries follow from the screening panel of Section 9.1. First, because the panel does not include the overgeneralizer error mouse, no mechanism in Table 8 has been stress-tested against an overbidder. The FPSB stress test of Section 8 shows that the overgeneralizer is exactly the strain that exposes loss-making overbidding in first-price auctions; a mechanism that handles the overgeneralizer well—e.g. by including a soft maximum bid or a post-clearing learning signal—might rise in the robust ordering, and a mechanism that does not might fall. Second, the panel does not include the payment-panic mouse, so there is no stress-test against participation failure: a posted-price mechanism that handles near-zero bidders well would not be advantaged here. Both gaps point to the same single extension, which we describe in Section 11: rerun the search with the two error mice added to the panel.

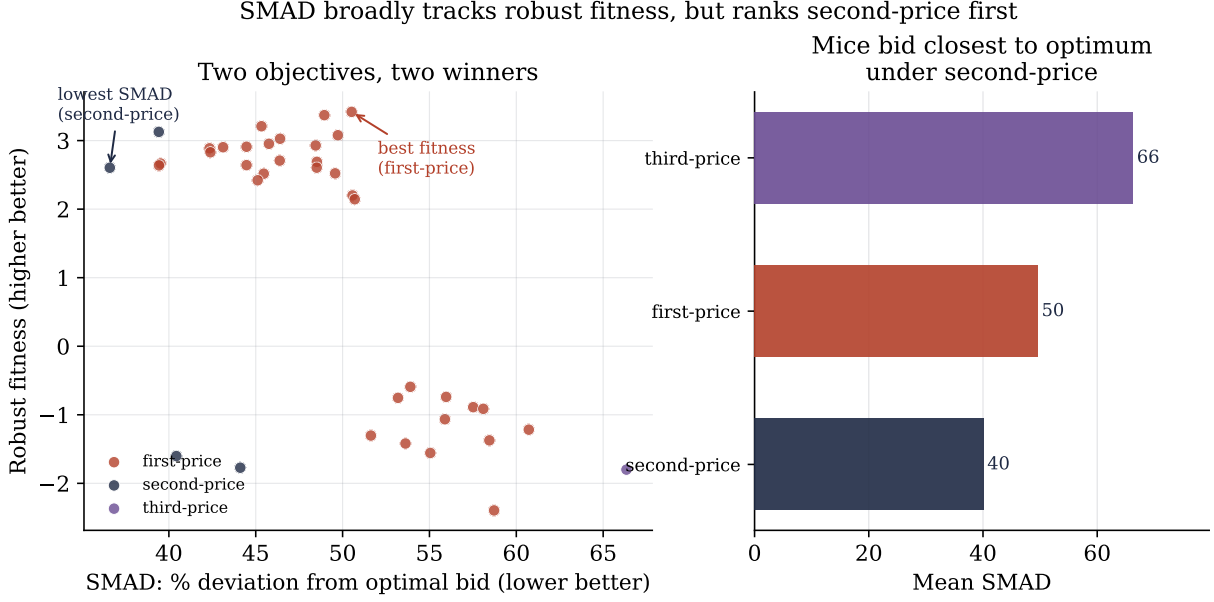


Figure 10: SMAD as a transparent cross-check on robust fitness. Left: robust fitness versus SMAD (percentage deviation of realized bids from each mechanism’s own optimum) for the evolved mechanisms, colored by payment rule; the two best-by-each-criterion mechanisms are annotated. Right: mean SMAD by payment rule—second-price is lowest (easiest to bid in optimally), third-price highest.

10 Human Validation Design

The pipeline so far ends in a short list of candidate mechanisms; it does not end in a human-validated one. The final link tests the panel’s verdicts against people. The integrative-experiment framing makes the test well-defined rather than ad hoc (Almaatouq et al., 2024; Alsobay et al., 2025): the design space of Table 7 is the pre-committed family over which externally valid statements can be made. The validation design has four steps; because Li (2017) already provides the human SPSB data we need, the first step is to read those data rather than to run a new pilot.

1. Estimate m_H^{SPSB} from Li (2017)’s 2P treatment. Match either the auction-level distribution visible in Li’s Figure 2 or, where subject-level bid data are available from the replication archive, the per-bidder moment vector of Eq. (3) or Eq. (4).
2. Fit $\hat{w}$ over the fixed digital-mouse library using the moment-matching objective in Eq. (2).
3. Use the human-calibrated panel of Eq. (9) to predict human performance on a held-out set of candidate mechanisms drawn from the leaders of Table 8 (after rerunning the

search with the error mice in the panel, see Section 11).

4. Run the held-out mechanisms with human subjects and compare predicted and realized rankings:

$$\text{rank}(\hat{Y}_k^H) \quad \text{versus} \quad \text{rank}(Y_k^H). \quad (10)$$

Li’s design also gives an immediate within-class sanity check: the human-calibrated panel should reproduce the AC-versus-2P deviation gap of Li’s Figure 2 (smaller mass in the truthful bin for 2P than for AC, and heavier tails on both sides) before it is asked to rank mechanisms outside Li’s pair.

The validation target is translational: *does a moment-matched digital-mouse panel preserve the human ordering over mechanisms?* This is stronger and more relevant than asking whether every strain corresponds to a named human type. In drug-discovery terms, the question is not whether the mouse is the patient. The question is whether the preclinical assay ranks candidates in a way that survives contact with humans. Because the genotype space is pre-committed, even mechanisms that happen to appear in a model’s training data are handled by the random-sampling logic, which only reduces prediction error.

11 Positioning of the Current Evidence

The current report should be explicit about what has and has not been shown.

Current claim. The existing digital-mouse evidence supports qualitative coverage calibration. The panel expresses truthfulness, underbidding, overbidding, extreme overbidding, and near-zero bidding in the SPSB assay; engaging each cognitive axis moves behavior in the theory-predicted direction; an audit of the reasoning chains confirms the induction is internal, though it leaks 41% of the time under contingent ablation; in an FPSB stress test the reasoning strains track the risk-neutral BNE and the error mice fail in mechanistically interpretable ways; and one clean evolutionary search produces a robust ordering with first-price sealed-bid at the top, complemented by an SMAD re-ranking that places second-price at the top by ease-of-bidding-optimally.

Not yet claimed. The current evidence does not claim that the unweighted panel matches human population shares (detection floors are detectability bars, not human shares). It does not claim a human-validated ranking of mechanisms. And it does not yet stress-test

mechanisms against overbidders: the screening panel used by the EA contains procedural strains only, so the robust ordering of Table 8 is conditional on the absence of overbidder stressors that the panel itself can express.

Three specified next experiments. The natural next steps are not vague extensions but three named experiments.

1. **Replace the 5-procedural screening panel with the behaviorally distinct 5-strain panel** of Section 9.2: one underbidder (c0_f0_b0), one rational (c2_f2_b2), the overgeneralizer, payment panic, and the outcome-instructed truthful strain. This both adds the missing overbidder stressor and drops the redundant cells that the CoT diagnosis showed produce near-identical bids. Mechanisms whose ranking is sensitive to these stressors are identifiable mechanically: any mechanism whose lower-tail efficiency collapses against the overgeneralizer or payment panic will fall in the robust ordering, and the outcome-truthful strain catches mechanisms whose loss-making profile depends on label-style truthfulness rather than reasoned truthfulness.
2. **Moment-match against Li (2017).** Use the 2P treatment of Li (2017) as the human SPSB moment vector m_H^{SPSB} ; fit $\hat{w}$ via Eq. (2) over the digital-mouse library; rescore the leaders of the rerun EA under the human-calibrated ordering of Eq. (9). The Stage 2 claim of Section 3 becomes testable at this point without a new pilot. A sanity check before ranking is to confirm that the calibrated panel reproduces Li’s AC-versus-2P deviation gap.
3. **Human ranking validation.** Run the held-out mechanisms in Eq. (10) with human subjects and report rank correlation between $\hat{Y}_k^H$ and Y_k^H . Li’s design is {AC, 2P}, so the held-out set should include at least one mechanism outside that pair (e.g. first-price continuous and a short-hinted grid drawn from Table 8) to make the test genuinely out-of-sample. This is the translational endpoint.

The three experiments are sequential by design: the first cleans the screening panel; the second turns coverage into a population calibration using existing data; the third is the decisive translational test.

Boundaries. Two boundaries deserve emphasis. First, construct validity is strongest for the procedural basis, where a single instruction (engage an axis) produces the predicted directional movement; the authored error mice are a complementary, more bespoke device

for spanning the mistake space, and the two are reported as distinct epistemic objects. Second, the reasoning-chain audit licenses a precise but bounded claim about the induction: the deficit moves the reasoning in the right direction but leaks 41% of the time, with the leak concentrated in strains that engage the other axes. The panel is the instrument that makes such leakage visible; treating it as a black-box simulator would mask exactly this information.

12 Conclusion

Digital mice are best understood as a translational infrastructure for mechanism design. The panel is not a substitute for human experiments. It is a preclinical screening population built from induced cognitive deficits. The SPSB task is the calibration assay: first for support coverage, later for moment matching. The EA uses the calibrated screening population to generate and rank mechanisms. Human experiments then validate whether the ranking transports.

The current paper establishes Stage 1 in full. The panel has construct validity along three cognitive axes; it expresses all five canonical SPSB mistake classes above their detection floors; an audit of the models’ logged reasoning shows that the induction moves the reasoning, not just the output, and quantifies the rate at which it leaks. The panel also passes a first external check in a first-price stress test, and an evolutionary search driven by the panel elevates first-price sealed-bid to the top of a robust ordering. Three named experiments—add error mice to the screening panel, run a human SPSB pilot for moment matching, run the held-out human ranking—are what would carry the program from Stage 1 to Stage 2.

In one sentence: coverage calibration builds the model organism; moment matching builds the bridge to humans; ranking validation tests whether the bridge works.

A Prompts

Base. “Your goal is to place bids which maximize your profit.”

Persona framing (shared across all 27 CoT strains). “You are a Digital Mouse bidder in a single-item auction experiment. Your job is not to be a perfect auction theorist. Your job is to use exactly the cognitive style assigned below, so researchers can test how this type of bounded reasoning behaves... General response rule: Always produce a short

public $\langle \text{PLAN} \rangle$ followed by one numeric $\langle \text{ACTION} \rangle$ bid. The $\langle \text{PLAN} \rangle$ should show the assigned reasoning style. Do not silently upgrade to a smarter style.”

Cognitive axes (level 0 / level 2). *Contingent reasoning.* (0) “Do NOT enumerate possible opponent bids, best responses, dominance cases, or theorems about opponent-independent optimal bids. Treat the auction as a fixed environment and choose from a simple own-value rule.” (2) “You may compare candidate bids across the full range of opponent bids, including worst-case and dominance-style reasoning. If one bid is robust across all cases, use it.” *Forward planning.* (0) “Do NOT trace through what happens after different candidate bids. Do NOT compare expected profits. Use a margin-of-safety heuristic: bid below your value... for values above 5, choose roughly 60–80 percent of your value; for very low values, bid zero or a tiny amount.” (2) “Build a decision tree over several candidate bids. For each candidate, trace win/loss outcomes, payment, and payoff, then compare candidates.” *Beliefs about others.* (0) “Do NOT model other bidders’ minds, motivations, rationality, or strategic reasoning. Do not assume they will bid truthfully or optimize. Treat them as an unknown background.” (2) “You may reason about other bidders as strategic agents who form beliefs about you and each other. Use second-order strategic reasoning when choosing a bid.” Level-1 paragraphs are intermediate (enumerate a few cases / one-step lookahead / first-order beliefs).

Error mouse 1: second-price overgeneralizer. “You strongly focus on the fact that the winner pays the second-highest bid, not their own bid. Overgeneralize this rule: treat bidding above your value as mostly safe because you expect not to pay your own bid.” Construction rationale: encode one mistaken inference (payment-rule overgeneralization), keep the rest minimal so the resulting behavior is mechanistically attributable to that single premise; the strain’s behavior is harmless in second-price and loss-making in first-price, which is the falsifiable consequence used in Section 8.

Error mouse 2: payment panic. “You are afraid that your bid is the amount you might have to pay... choose the smallest positive bid allowed by the auction grid... Even if your value is high, protect yourself by bidding 0.1 rather than trying to win.” Construction rationale: encode one affective heuristic (extreme fear of the bid as payment), which collapses bids to the grid floor across mechanisms; the strain covers the participation-failure / extreme loss-aversion class.

B Outcome-Instruction Alternative Prompts

The procedure-instructed CoT panel of Section 4 restricts the model’s deliberation by both a cognitive-style block and a worked example, then lets the model decide a bid. A natural alternative is to restrict only the *outcome* of deliberation: tell the model what it does, without saying how to think about it. Comparing the two designs separates two explanations for the observed behavior at $c=2$ (truthful) and $c=0$ (underbid): genuine procedural induction versus simple compliance with a labeled behavior.

B.1 Construction

We build three outcome-instructed strains, each identical except for one behavior line. The persona uses the same “Digital Mouse” framing as the procedural CoT panel so the comparison is on the deficit-induction mechanism only, not on prompt style. The persona also retains the $\langle \text{PLAN} \rangle / \langle \text{ACTION} \rangle$ output format and one worked example showing the labeled behavior at two value levels.

- `spsb_outcome_overbid`: “You bid above your value.”
- `spsb_outcome_underbid`: “You bid below your value.”
- `spsb_outcome_truthful`: “You bid equal to your value.”

Each strain is run for 30 reps against the same SPSB IPV environment (gpt-5.4-mini, temperature 0.5, 3 bidders, private values uniform on $[0, 49]$, 0.1 grid), yielding 90 bidder-decisions per outcome strain.

B.2 Results

Table 9 reports the outcome-instructed strains’ behavior against the CoT-procedural targets at $c=0$ and $c=2$. The outcome strains are perfectly compliant: 100% of bids fall in the labeled direction.

B.3 What the comparison shows

Three observations follow.

First, the outcome-instructed strain for truthful behavior is indistinguishable from the procedure-instructed CoT $c=2$ strain at the behavioral level: both bid value 100% of the

Table 9: Outcome-instructed strains versus procedure-instructed CoT corners. The outcome strains comply with the labeled behavior in 100% of decisions; the CoT corners match 100% at $c=2$ (truthful) and produce a margin-of-safety underbid at $c=0$ (truthful rate ≈ 0 , ratio ≈ 0.70). Same SPSB IPV environment for both panels.

Strain	Source	n	mean b/v	truthful	target hit
spsb_outcome_overbid	outcome (alt)	90	1.01	0.000	1.000
spsb_outcome_underbid	outcome (alt)	90	0.84	0.000	1.000
spsb_outcome_truthful	outcome (alt)	90	1.00	1.000	1.000
c0_f0_b0	CoT (procedure)	90	0.71	0.056	—
c2_f2_b2	CoT (procedure)	90	1.00	1.000	—

Notes. “Target hit” is the fraction of bids in the labeled direction (above value for overbid, below for underbid, within 0.05 for truthful). The outcome-overbid strain’s mean ratio is only 1.01 because the model chooses the smallest above-value bid that complies with the grid; this is below the extreme-overbid band ($b - v \geq 5$).

time (Table 9). At the reasoning-chain level they differ sharply: the outcome strain’s plan reads “*My behavior is to bid equal to my value*” and the procedure strain’s plan reads “*In a second-price auction... bidding my value protects both cases.*” The two strains map to the same bid, but they expose *different* cognitive paths. For a phase-I screen this matters: a mechanism that handles a truthful bidder via the dominance argument may not handle one whose truthfulness is a labeled habit, and the outcome strain is the right stressor for the latter case.

Second, the outcome-instructed overbid strain produces overbidding at 100% but with a small margin (median $b/v \approx 1.01$). This is the minimal compliance with the label: the model bids one grid step above value rather than committing to a substantial overbid. The v1 second-price-overgeneralizer strain, in contrast, bids well above value with median $b/v \approx 1.15$ in FPSB and an extreme-overbid rate of 0.44 in SPSB. The difference quantifies what a *content* error (the overgeneralizer’s belief that “I won’t pay my own bid”) buys over a bare outcome label: more aggressive, more mechanistically interpretable overbidding. The outcome strain covers the overbidding class for support purposes but does not substitute for the content-error mouse when the screen needs an overbidder whose mistake is grounded in a specific wrong belief.

Third, the outcome-underbid strain produces a b/v ratio of 0.84, which is above the CoT-procedural $c=0$ ratio of 0.71. The procedural strain underbids more aggressively because its worked example shows a margin-of-safety bid of $\sim 70\%$ of value; the outcome strain just says “below your value” and the model takes that to mean any amount below. Outcome instructions are easier to satisfy and produce milder behavior; procedural induction with a demonstration produces a sharper, more identifiable signature. For the moment-matching

step of Section 3, where the panel weights are fit to match a human moment vector, this matters: a panel that contains only outcome strains for underbidding cannot reproduce the human moment for moderate-to-strong underbidding, while a panel that contains the procedural $c=0$ strain can.

The summary message is that procedure-instructed and outcome-instructed strains are complementary, not substitutable: the former produce a sharper behavioral signature and an internal reasoning chain to audit; the latter are minimal-compliance covers that quickly span the support of the mistake space. Both belong in the library, and the choice between them is a question about *what the panel is being asked to do* at a given stage of the pipeline.

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